

hw1

chl

英文教材：p20 Ex.6, 9

中文参考书（概率论教程-中科大出版社）：p37, Ex. 8,9,10.

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6.

The union of a countable collection of B.F.'s $\{\mathcal{F}_j\}$ such that $\mathcal{F}_j \subset \mathcal{F}_{j+1}$ need not be a B.F., but there is a minimal B.F. containing all of them, denoted by $\bigvee_j \mathcal{F}_j$. In general $\bigvee_{\alpha \in A} \mathcal{F}_\alpha$ denotes the minimal B.F. containing all $\mathcal{F}_\alpha$, $\alpha \in A$. [HINT: Ω = the set of positive integers; $\mathcal{F}_j$ = the B.F. generated by those up to j .]

Proof. Consider $\Omega = \bigcup_{n=1}^{\infty} \{n\}$, we define

$$\mathfrak{F}_j = \sigma(\{n\}, n \geq 1) \triangleq \{X, X \cup \{j+1, j+2, \dots\} : X \in 2^{\{1,2,\dots,j\}}\} \subset 2^\Omega$$

We write $T_j \triangleq \{j+1, j+2, \dots\}$, $K_j \triangleq \{1, 2, \dots, j\}$,

1. $\forall 1 \leq k \leq j, \{k\} \in \mathfrak{F}_j$.
2. $\Omega = K_j \cup T_j \in \mathfrak{F}_j$.
3. $\forall A \in \mathfrak{F}_j$. If $A = B \cup T_j$, $B \subset K_j$, $A^c = K_j \setminus B \in \mathfrak{F}_j$; if $A \subset K_j$, $A^c = (K_j \setminus A) \cup T_j \in \mathfrak{F}_j$.

4. $\{A_n = B_n \cup T_j \text{ or } B_n\}_{n=1}^\infty \in \mathfrak{F}_j, B_n \in K_j,$

$$\bigcup_{n=1}^\infty A_n = \left(\bigcup_{n=1}^\infty B_n \right) \cup T_j \text{ or } \left(\bigcup_{n=1}^\infty B_n \right) \in \mathfrak{F}_j$$

5. $\forall A \in \mathfrak{F}_j$. If $A = B \cup T_j = (B \cup \{j+1\}) \cup T_{j+1}, B \subset K_j, B \cup \{j+1\} \subset K_{j+1}, A \in \mathfrak{F}_{j+1};$
if $A \subset K_j \subset K_{j+1}, A \in \mathfrak{F}_{j+1}.$

Therefore, $\{\mathfrak{F}_i\}$ is a countable collection of B.F.'s such that $\mathfrak{F}_j \subset \mathfrak{F}_{j+1}$. However, $\bigcup_{n=1}^\infty \mathfrak{F}_j = \{A \in 2^\Omega : |A| < \infty \text{ or } |A^c| < \infty\}$ does not contain $\{2n : n \in \mathbb{N}\}$. However, $\forall n \in \mathbb{N}, 2n \in \Omega$, any B.F. containing Ω must contain $\bigcup_{n=1}^\infty \{2n\} = \{2n : n \in \mathbb{N}\}.$

Considering

$$\mathcal{G} = \{A : A \text{ is B.F. and } \mathfrak{F}_j \subset A, \forall j\}$$

We know that $2^\Omega \in \mathcal{G}$, so $\mathcal{G}$ is non-empty, then we can say

$$\bigvee_{j=1}^\infty \mathfrak{F}_j = \bigcap_{A \in \mathcal{G}} A$$

□

9.

If $\mathcal{F}$ is a B.F. generated by a countable collection of disjoint sets $\{\Lambda_n\}$, such that $\bigcup_n \Lambda_n = \Omega$, then each member of $\mathcal{F}$ is just the union of a countable subcollection of these Λ_n 's.

Proof. Since for any B.F. $\mathcal{G}$ containing $\{\Lambda_n\}$, $\forall I \subset \mathbb{N}, \bigcup_{i \in I} \Lambda_i \in \mathcal{G}$, we obtain

$$\mathcal{G}_0 \triangleq \left\{ \bigcup_{i \in I} \Lambda_i : I \subset \mathbb{N} \right\} \subset \bigcap_{\{\Lambda_i\}_{i \in I} \text{ is B.F.}} \mathcal{G}$$

We next prove $\mathcal{G}_0$ is B.F.

1. $\Omega = \bigcup_n \Lambda_n \in \mathcal{G}_0$.
2. $\forall A = \bigcup_{i \in I \subset \mathbb{N}} \Lambda_i \in \mathcal{G}_0, A^c = \bigcup_{i \in \mathbb{N} \setminus I \subset \mathbb{N}} \Lambda_i \in \mathcal{G}_0$.
3. $\forall A_n = \bigcup_{i \in I_n \subset \mathbb{N}} \Lambda_i \in \mathcal{G}_0, \bigcup_n A_n = \bigcup_{i \in \bigcup_n I_n \subset \mathbb{N}} \Lambda_i \in \mathcal{G}_0$

Therefore

$$\mathcal{G}_0 \triangleq \left\{ \bigcup_{i \in I} \Lambda_i : I \subset \mathbb{N} \right\} = \bigcap_{\mathcal{G} \supset \{\Lambda_i\}_i, \mathcal{G} \text{ is B.F.}} \mathcal{G}$$

□

8.

设 $\mathcal{C}$ 和 $\mathcal{F}$ 是两个集类, 且 $\mathcal{C} \subseteq \mathcal{F}$, 证明:

- (1) 若 $\mathcal{C}$ 为代数, 且 $\mathcal{F}$ 为单调类, 则 $\sigma(\mathcal{C}) \subseteq \mathcal{F}$.
- (2) 若 $\mathcal{C}$ 为 π 系, 且 $\mathcal{F}$ 为 λ 系, 则 $\sigma(\mathcal{C}) \subseteq \mathcal{F}$.

(1)

Proof. $\mathcal{C}$ 为代数, 则

$$A \in \mathcal{C} \implies A^c \in m(\mathcal{C}); \quad A, B \in \mathcal{C} \implies A \cap B \in m(\mathcal{C}).$$

由 9. 定理得

$$\sigma(\mathcal{C}) = m(\mathcal{C}) \subset \mathfrak{F}$$

□

(2)

Proof. $\mathcal{C}$ 是 π 系, 则

$$A, B \in \mathcal{C} \implies A \cap B \in \lambda(\mathcal{C}).$$

由 9. 定理得

$$\sigma(\mathcal{C}) = \lambda(\mathcal{C}) \subset \mathfrak{F}$$

□

9.

设 $\mathcal{C}$ 是一个集类, 证明:

(1) $m(\mathcal{C}) = \sigma(\mathcal{C})$ 当且仅当

$$A \in \mathcal{C} \implies A^c \in m(\mathcal{C}); \quad A, B \in \mathcal{C} \implies A \cap B \in m(\mathcal{C}).$$

(2) $\lambda(\mathcal{C}) = \sigma(\mathcal{C})$ 当且仅当

$$A, B \in \mathcal{C} \implies A \cap B \in \lambda(\mathcal{C}).$$

(1)

Proof. 显然有 $m(\mathcal{C}) \subset \sigma(\mathcal{C})$, 故只需证 $m(\mathcal{C}) \supset \sigma(\mathcal{C})$, 等价于证明 $m(\mathcal{C})$ 是 σ -域, 要求分别满足

$$1. \forall A, B \in m(\mathcal{C}) \implies AB \in m(\mathcal{C})$$

$$\iff \mathcal{G}_1 \triangleq \{A \in m(\mathcal{C}) \mid \forall B \in m(\mathcal{C}), AB \in m(\mathcal{C})\} = m(\mathcal{C})$$

显然 $\mathcal{G}_1$ 是单调类, 因此只需证 $\mathcal{G}_1 \supset \mathcal{C}$,

$$\iff \mathcal{G}_2 \triangleq \{A \in m(\mathcal{C}) \mid \forall B \in \mathcal{C}, AB \in m(\mathcal{C})\} = m(\mathcal{C})$$

显然 $\mathcal{G}_2$ 是单调类, 因此只需证 $\mathcal{G}_2 \supset \mathcal{C}$, 即 $\forall A \in \mathcal{C}, \forall B \in \mathcal{C}, AB \in m(\mathcal{C})$, 这是条件 2.

$$2. \forall A \in m(\mathcal{C}), A^c \in m(\mathcal{C})$$

$$\Longleftrightarrow \mathcal{G}_3 \triangleq \{A \in m(\mathcal{C}) \mid A^c \in m(\mathcal{C})\} = m(\mathcal{C})$$

显然 $\mathcal{G}_3$ 是单调类，因此只需证 $\mathcal{G}_3 \supset \mathcal{C}$,

$$\Longleftrightarrow \mathcal{G}_4 \triangleq \{A \in \mathcal{C} \mid A^c \in m(\mathcal{C})\} = \mathcal{C}$$

这是条件 1.

$$3. \forall A_n \in m(\mathcal{C}), \text{ 根据单调类定义和已证结论有}$$

$$\bigcup_{k=1}^n A_k \in m(\mathcal{C}) \quad \bigcup_{n=1}^{\infty} A_n = \bigcup_{n=1}^{\infty} \bigcup_{k=1}^n A_n \in m(\mathcal{C})$$

□

(2)

Proof. 显然有 $\lambda(\mathcal{C}) \subset \sigma(\mathcal{C})$, 故只需证 $\lambda(\mathcal{C}) \supset \sigma(\mathcal{C})$, 等价于证明 $\lambda(\mathcal{C})$ 是 σ -域, 要求分别满足

$$1. \forall A, B \in \lambda(\mathcal{C}) \implies AB \in \lambda(\mathcal{C})$$

$$\Longleftrightarrow \mathcal{G}_5 \triangleq \{A \in \lambda(\mathcal{C}) \mid \forall B \in \lambda(\mathcal{C}), AB \in \lambda(\mathcal{C})\} = \lambda(\mathcal{C})$$

显然 $\mathcal{G}_5$ 是 λ 系, 只需证 $\mathcal{G}_5 \supset \mathcal{C}$,

$$\Longleftrightarrow \mathcal{G}_6 \triangleq \{A \in \lambda(\mathcal{C}) \mid \forall B \in \mathcal{C}, AB \in \lambda(\mathcal{C})\} = \lambda(\mathcal{C})$$

显然 $\mathcal{G}_4$ 是 λ 系, 只需证 $\mathcal{G}_6 \supset \mathcal{C}$, 即 $\forall A, B \in \mathcal{C}, AB \in \lambda(\mathcal{C})$, 这是条件.

$$2. \forall A \in \lambda(\mathcal{C}), A^c \in \lambda(\mathcal{C}), \text{ 这由 } \lambda \text{ 系的定义直接给出.}$$

$$3. \forall A_n \in \lambda(\mathcal{C}), A_n \setminus \bigcup_{k=1}^{n-1} A_k \in \lambda(\mathcal{C}),$$

$$\bigcup_{n=1}^{\infty} A_n = \bigcup_{n=1}^{\infty} \left(A_n \setminus \bigcup_{k=1}^{n-1} A_k \right) \in \lambda(\mathcal{C})$$

□

10.

设 $\mathcal{C}$ 是一个集类, 证明:

(1) $m(\mathcal{C}) = \sigma(\mathcal{C})$ 当且仅当

$$A \in \mathcal{C} \implies A^c \in m(\mathcal{C}); \quad A, B \in \mathcal{C} \implies A \cup B \in m(\mathcal{C}).$$

(2) $\lambda(\mathcal{C}) = \sigma(\mathcal{C})$ 当且仅当

$$A, B \in \mathcal{C} \implies A \cup B \in \lambda(\mathcal{C}).$$

(1)

Proof. 由上题知 $m(\mathcal{C}) = \sigma(\mathcal{C})$ 当且仅当

$$A \in \mathcal{C} \implies A^c \in m(\mathcal{C}); \quad A, B \in \mathcal{C} \implies A \cap B \in m(\mathcal{C})$$

当且仅当

$$A \in \mathcal{C} \implies A^c \in m(\mathcal{C}); \quad A, B \in \mathcal{C} \implies A \cup B \in m(\mathcal{C})$$

这是因为

$$\forall A, B \in m(\mathcal{C}), A^c, B^c \in m(\mathcal{C}), A^c \cup B^c \in m(\mathcal{C}), A \cap B = (A^c \cup B^c)^c \in \mathcal{C}.$$

$$\forall A, B \in m(\mathcal{C}), A^c, B^c \in m(\mathcal{C}), A^c \cap B^c \in m(\mathcal{C}), A \cup B = (A^c \cap B^c)^c \in \mathcal{C}. \quad \square$$

(2)

Proof. 由上题知 $\lambda(\mathcal{C}) = \sigma(\mathcal{C})$ 当且仅当

$$A, B \in \mathcal{C} \implies A \cap B \in \lambda(\mathcal{C})$$

当且仅当

$$A, B \in \mathcal{C} \implies A \cup B \in \lambda(\mathcal{C})$$

这是因为

$A, B \in \mathcal{C} \subset \lambda(\mathcal{C}), A^c, B^c \in \lambda(\mathcal{C}), A^c \cup B^c \in \lambda(\mathcal{C}), A \cap B = (A^c \cup B^c)^c \in \lambda(\mathcal{C}).$
 $A, B \in \mathcal{C} \subset \lambda(\mathcal{C}), A^c, B^c \in \lambda(\mathcal{C}), A^c \cap B^c \in \lambda(\mathcal{C}), A \cup B = (A^c \cap B^c)^c \in \lambda(\mathcal{C}).$

□